Questions:

1. **Which product line was the most profitable? Report the summary statistics for each distribution, and comment.**

   (Appendix 1.1) Historically speaking, the most profitable product line is GTMSys, which has made 844.07 billion dollars for our company. The average profit of GTMSys is lower than the average profits generated by all our products we put into the market (3.72 vs. 3.75), and the range is shorter than the average, indicating fewer outliers and lower volatility. And with the exceptionally high number of opportunities (227 out of 481), almost half of our products that won the bid were GTMSys. GTMSys is the most profitable product for us. Statistics also show that GTMSys has a moderate stdDev (1.17), a small distance from median to average (1.18), and a perfect match between mean and median (3.72). All these mean that GTMSys is a very mature product, it has gone through the test of “times”. In a nutshell, GTMSys is our money tree.

2. **How does the total profit of each market break down by product? Create an appropriate visual aid showing each product’s relative share of profits and comment.**

   (Appendix 2.1) Our teams are selling GTMSys very well, basically everywhere, except India and Canada. Procsys is popular too, in some of these countries. Finsys, LearnSys, Logissys, Lifesys, and ContactSys are gradually exploring the market. In the UK, our main market, product diversity is high, and each of our products holds a share of the UK market. In Spain and Singapore, all the products they purchased from us are GTMSys. Other European markets have demands for Finsys, GTMSys, and Lifesys. GTMSys accounted for a significant proportion of our sales in Japan, the Americas, and Africa. Japan and the Americas also like Procsys, while in Africa, the rest of the market is filled with learnSys. In India and Canada, our customers only purchase Procsys.

3. **Which markets were most valuable? Report summary statistics for market-specific revenues and comment. Do the same with profits. (Football Field Graph)**

To identify the most valuable market, the summary statistics must be interpreted jointly. A high mean with large size indicates a solid market with frequent, sizable deals, while small size but high mean reflects riskier, limited opportunities. High volatility signals uncertainty, and the mean–median gap shows skewness in deal size.

Revenue analysis (Appendix 3.1): Japan, Singapore, Canada and India show the highest average revenues, all above 8.19M. However, smaller samples suggest premium and niche markets. The UK stands out with a strong mean (8.15M), large volume (N=553), and low volatility (SD=2.01), making it the most valuable market. In most regions, median align with mean. Singapore and Spain show lower medians, suggesting a few large deals distort the averages. Japan, India and Singapore show high standard deviations, but also high returns. Canada, on the other side, requires further testing for data reliability because of the small sample size.

Profit analysis (Appendix 3.2): Profitability reflects tax rate, tariff, and policy impacts. The UK remains the most valuable market. It persisted with strong mean profit, low volatility, and fewer extremes. Singapore and Japan showed rightward skewness, making them less attractive markets. The Americas exhibit a left skewed distribution. This indicates possible small outliers dragging down the mean. This might suggest undervaluation. Other Europe is balanced for both revenue and profit. Although it has lower average profit, its volume (N=158) still makes it a relatively valuable market.

Successful deals analysis (Appendix 3.3 and 3.4): The earlier analysis included all deals, but only focusing on the company's wins gives a clearer picture. Incorporating the BCG matrix would also provide deeper evaluation. In the UK, profit is below the average, but the company controls nearly half the market. It is still the biggest and safest market with a lot of potential. The Americas and other Europe remain stable, with the potential to raise profit through cost management. Japan and India show higher than average revenue and near average profits. The company has strong control in both markets, making them strategically vital. Africa emerges as a reliable median size market. The company has strong market share in Africa and is performing at average. Africa’s low standard deviation shows it is a stable market. Singapore, however, shows below average revenue and profit levels with low market share. It is unattractive unless the company could effectively expand market share.

Overall, the UK stands out as the most valuable market. This is because it is the largest market and nearly half of it is under company control. Company's below average profit signals upside potential. The company could grow profitability through enhancing current market control and expanding future markets. Therefore, it would be identified as a **Cash Cow** with the potential of being a **Star**. The Americas, Africa, and Other Europe are **Cash Cows** because of their strong volumes, and stable performance. They also show room for possible profit improvements. Although Japan and India are niche markets, they provide high revenue and profit. These markets are **Stars** and require continued investment for expansion. Singapore shows lower revenue, weaker profitability and small market share. It is classified as a **Dog**. This only changes if the company could steal market share from its competitors. Overall, the UK, Japan, and India stand out as the most valuable markets.

4. How does the probability of conversion vary across markets? Represent this visually.

(Appendix 4.1) Win or loss are represented by the binary variable "SalesOutcome", with 1 representing a win and 0 as a loss. Conversion rate by market highlights performance differences.

Canada (67%, N=6) and Japan (62%, N=16) are the highest conversion markets. However, both are niche markets with small sample sizes, especially Canada. Africa (59%, N=93) and the Americas (53%, N=104) also show high conversion and large volumes.

India (49%, N=35) and the UK (48%, N=553) sit near the average. The UK stands out because of its very large size. Other Europe (42%, N=158), the firm has slightly weaker controls, but still holds value through its volume.

Singapore (26%, N=23) and Spain(8%, N=12) have the lowest conversion rates. These markets appear unattractive, but this might be biased from small sample size.

Visual Comparison shows clearly the level of conversion rate among markets. Conversion rate results prove the previously identified valuable markets. By contrast, Singapore and Spain look even less attractive, with very low conversion rates.

5. **Is the size of a customer's profits related to the size of the deals with WSES? Make a scatter plot and discuss any relationship or lack thereof.**

   There is no linear relationship between customer's profit and size of the deals. The scatter plot (Appendix 5.1) shows no significant pattern. By calculating the correlation coefficient (Appendix 5.2), we can find out that r = -0.01 and p-value = 0.744, indicating that there is no linear relationship between two variables.

6. **Compare the Indian and UK markets in terms of average deal size, profitability, customer size, and probability of conversion. Are they "distinct" statistically speaking? Compare the markets along each dimension with a t-test and comment.**

   After doing the T-test for these 4 variables, we can conclude that there were no statistically differences between two markets. The average deal size was similar, with p-value = 0.9347(Appendix 6.1). Neither profitability(Appendix 6.2) nor customer size (Appendix 6.3) showed significant difference, both of the p-values are greater than 0.05. Conversion rates(Appendix 6.4) were almost identical(India = 48.6%, UK = 48.3%, p-value = 0.97).

7. **Which markets should WSES prioritize? Should they focus on certain products in certain markets? Make a good argument supported by data and sound reasoning.**

   7-1: Which markets should WSES prioritize?

   To determine which markets WSES should prioritize, we first analyzed the relationship between region and sales outcome using a chi-square test (Appendix 7.1). The result was statistically significant (p = 0.004), indicating that conversion rates vary meaningfully across regions.

   In addition to sales outcome, we also considered whether average sales value differs significantly across regions, as this could offer another basis for prioritization. However, an ANOVA test revealed no statistically significant difference in average sales value by region ($p = 0.261$; Appendix 7.2). This indicates that deal size is relatively consistent across markets, making conversion rate a more reliable metric for decision-making.

As shown in Appendix 7.3, Africa stands out with both a high conversion rate and a sufficient sample size, making it the most reliable priority market. Canada and Japan also show promising potential due to their strong performance; however, their small sample sizes warrant cautious interpretation and further investigation. As for the UK and Other Europe, although WSES has invested considerable effort in these regions, the markets appear highly competitive, and their success rates fall below 50%. Therefore, WSES should prioritize Africa as the most strategically sound market.

7-2: Should WSES focus on certain products in certain markets?

To evaluate whether WSES should focus on certain products in certain markets, we first examined whether sales outcomes vary by product type. A chi-square test confirmed a statistically significant association between product type and sales outcome ($p < 0.001$; Appendix 7.4), indicating that conversion rates differ meaningfully across products.

Building on this result, we further analyzed win rates across product-region combinations (Appendix 7.5). The results reveal that certain products perform better in specific markets.

These findings strongly support the recommendation that WSES should focus on certain products in certain markets. Specifically, GTMSys has shown a high conversion rate in Africa (61.3%) and the Americas (54.9%), making these regions high-priority targets for this product. LearnSys and Lifesys also perform well in both Africa and the UK, suggesting strong product-market fit and room for expansion. Logissys, with a high conversion rate in the UK (69.0%), represents another valuable opportunity.

In contrast, Finsys has underperformed across all observed regions, particularly in the UK (24.1%), and may require a complete strategic reassessment or even phase-out. Similarly, ContactSys (UK only) and GTMSys in Spain (8.3%) showed weak results and should be deprioritized.

8. **Should WSES focus on joint ventures in the future? Compare the probability of conversion when WSES has more than 50% of a project (vs less than 50%) using an appropriate test and comment.**

   To assess whether WSES should focus on joint ventures in the future, we conducted a two-sample z-test comparing the conversion rates of projects where WSES holds a majority stake (more than 50%) versus those where it holds 50% or less.

   The results (Appendix 8.1) show that projects with WSES ownership above 50% achieve a significantly higher conversion rate (mean = 49.8%) compared to those with lower ownership (mean = 39.6%). This approximately 10-percentage-point difference is statistically significant at the 5% level (p = 0.016).

   These findings suggest that WSES is more likely to win bids when it retains a controlling interest in joint ventures. Therefore, WSES should prioritize opportunities where it can maintain a majority share and adopt a more selective approach for bids where its ownership is limited.

9. **Are the "hunches" of the sales team any good? Divide all leads into "E/V" and "F/L", compare their success in sales conversion using an appropriate test, and comment.**

   Since both variables, sales outcome (Won/Lost) and lead conversion class (E/V vs F/L), are categorical and independent, we conducted a Chi-square test of independence, shown in Appendix 9. The test gave a p-value of 0.16, which is higher than the 0.05 significance level. This indicates that there is no statistically significant relationship between the sales team's confidence classification and the actual deal outcome. Although the high-confidence group (E/V) achieved a slightly higher success rate than the low-confidence group (F/L), the difference is small and may have occurred by chance, so the hunches are not reliable predictors of sales success.

10. **Is there any relationship between marketing expenses (6% of sales revenue) and WSES' profitability on a deal? Make a scatter plot of the two variables and comment.**

As shown in Appendix 10, the scatter plot illustrates the relationship between marketing expenses (6% of sales revenue) and WESE's profitability (profit percentage multiplied by sales revenue). In this analysis, only sales outcomes equal to 1 were included, meaning that only successful transactions were considered, while rows with a sales outcome of 0 were excluded. The data points display a clear upward trend, the yellow trendline shows a noticeable upward slope, and the calculated correlation coefficient (r = 0.7474) indicates a strong positive linear relationship between marketing expenses and profitability. In other words, increased marketing expenses tend to enhance the overall profitability.

11. Comment on other aspects of the firm’s approach which you think might be important for them to focus on as they try to grow the business.

As WSES cannot easily switch to other countries or products in the short term, they should strengthen its internal strategy to support sustainable growth. The company needs to shift from instinct driven decision making to a data driven model. They must apply analytics to target winnable, high value leads. Data driven models not only strengthen market choosing decisions, but also help management identify key initiatives that boost profitability. Statistics would strengthen their cost reduction strategies, such as reevaluating underperforming products, and supporting successful product lines. WSES should also focus on amplifying its influence in joint ventures, as evidence shows higher success when the firm holds majority control.

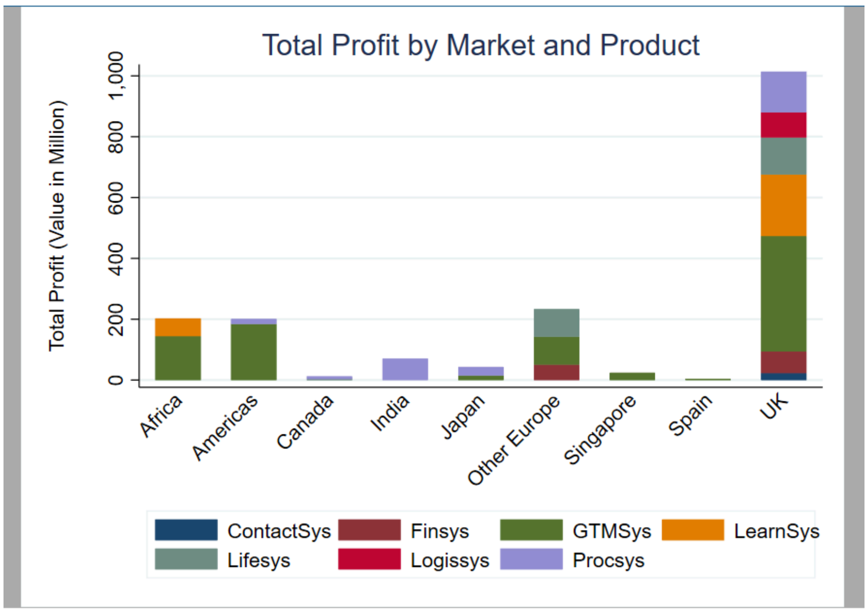
As WSES focuses on BSB clients, it is vital to keep customer relationships. BSB clients value reliability and long term engagement. Building stronger customer service and customization, supported by software solutions like CRM systems, can enhance retention and open new business opportunities.

Appendix:

Appendix 1.1 Summary Statistics of Product Line Performance

Product	sum	mean	p50	sd	min	max	N
ContactSys	22.95	3.82	4.16	1.03	2.56	5.11	6.00
Finsys	121.71	3.58	3.43	1.15	1.54	6.02	34.00
GTMSys	844.03	3.72	3.72	1.17	0.71	6.67	227.00
LearnSys	260.53	3.67	3.71	0.97	1.67	6.58	71.00
Lifesys	215.92	4.00	3.98	1.28	1.35	7.99	54.00
Logissys	82.40	4.12	4.11	1.13	2.39	6.17	20.00
Procsys	256.69	3.72	3.58	1.39	1.14	7.68	69.00
Total	1804.23	3.75	3.74	1.18	0.71	7.99	481.00

Appendix 2.1 Stacked Bar Chart of Regional Profit Contributors



Appendix 3.1. Summary Statistics of Whole Market's Sales Revenue (Millions)

Region	Mean	p50	SD	Min	Max	N
Africa	7.82	7.69	1.89	4.02	13.18	93.00
Americas	7.76	7.69	1.89	3.07	12.28	104.00
Canada	8.19	8.30	1.27	6.87	9.43	6.00
India	8.19	8.00	2.31	3.75	14.23	35.00
Japan	8.90	8.83	2.01	5.24	12.34	16.00
Other Europe	7.84	7.75	1.88	1.64	12.23	158.00
Singapore	8.20	7.90	2.18	4.56	12.59	23.00
Spain	7.96	7.61	2.02	4.30	11.45	12.00
UK	8.15	8.14	2.01	2.40	14.01	553.00
Total	8.04	8.00	1.98	1.64	14.23	1000.00

Appendix 3.2. Summary Statistics of Whole Market's Profit (Millions)

Region	Mean	p50	SD	Min	Max	N
Africa	3.97	3.93	1.28	1.46	8.17	93.00
Americas	3.89	4.00	1.15	1.32	6.40	104.00
Canada	3.54	3.27	1.14	2.34	5.69	6.00
India	4.38	4.14	1.55	1.80	7.70	35.00
Japan	4.33	3.97	1.47	2.25	6.66	16.00
Other Europe	3.95	3.73	1.29	0.71	7.56	158.00
Singapore	4.51	4.09	1.59	2.20	8.44	23.00
Spain	4.17	4.04	1.19	2.49	6.64	12.00
UK	4.12	4.02	1.30	1.14	8.54	553.00
Total	4.08	3.97	1.30	0.71	8.54	1000.00

Appendix 3.3. Summary Statistics of Company's Sales Revenue (Millions)

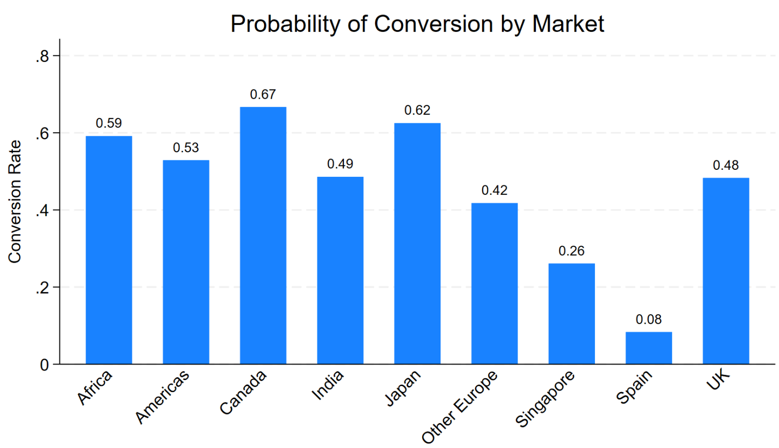
Region	Mean	p50	SD	Min	Max	N
Africa	7.69	7.46	1.82	4.17	12.51	55.00
Americas	7.87	7.73	1.84	3.74	11.60	55.00
Canada	8.11	8.06	1.43	6.87	9.43	4.00
India	8.53	8.33	2.60	3.75	14.23	17.00
Japan	9.01	8.95	2.15	5.24	12.34	10.00
Other Europe	7.76	7.88	2.07	1.64	12.23	66.00
Singapore	8.52	8.65	2.27	4.56	10.99	6.00
Spain	7.76	7.76	.	7.76	7.76	1.00
UK	8.24	8.30	1.99	2.40	14.01	267.00
Total	8.10	8.07	1.99	1.64	14.23	481.00

Appendix 3.4. Summary Statistics of Company's Profit Value (Millions)

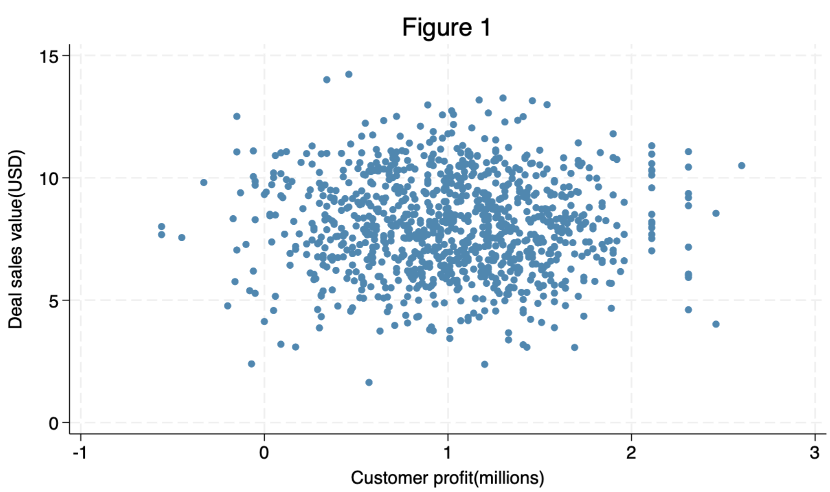
Region	Mean	p50	SD	Min	Max	N
Africa	3.69	3.73	1.14	1.46	6.61	55.00
Americas	3.65	3.89	1.12	1.40	5.78	55.00
Canada	2.99	3.12	0.46	2.34	3.39	4.00
India	4.13	3.67	1.59	1.80	7.68	17.00
Japan	4.30	3.97	1.52	2.25	6.66	10.00
Other Europe	3.53	3.39	1.18	0.71	7.37	66.00
Singapore	4.00	3.55	1.60	2.23	6.48	6.00
Spain	4.27	4.27	.	4.27	4.27	1.00
UK	3.80	3.80	1.15	1.14	7.99	267.00
Total	3.75	3.74	1.18	0.71	7.99	481.00

Appendix 4.1. Conversion Probability Table and Bar Chart

Region	Mean	N
Africa	0.59	93.00
Americas	0.53	104.00
Canada	0.67	6.00
India	0.49	35.00
Japan	0.62	16.00
Other Europe	0.42	158.00
Singapore	0.26	23.00
Spain	0.08	12.00
UK	0.48	553.00
Total	0.48	1000.00



Appendix 5.1 Relationship Between Customer Profit and Deal Sales Value



Appendix 5.2. Correlation Output between Sales Value and Customers' Profit

```
. pwcorr SalesValue ProfitofCustomer, sig

             | SalesV~n Profit~n
-------------+------------------
SalesValue~n |   1.0000
             |
             |
ProfitofCu~n |  -0.0103   1.0000
             |   0.7439
             |
```

Appendix 6.1. T-test of Average Deal Size

```
. ttest SalesValue, by (Region) unequal

Two-sample t test with unequal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
   India |      35    8.186857    .3903706    2.309464    7.393529    8.980186
      UK |     553     8.15387    .0855969    2.012894    7.985734    8.322005
---------+--------------------------------------------------------------------
Combined |     588    8.155833     .083698    2.029569    7.991449    8.320217
---------+--------------------------------------------------------------------
    diff |            .0329873    .3996449               -.7765194     .842494
------------------------------------------------------------------------------
    diff = mean(India) - mean(UK)                                 t =   0.0825
H0: diff = 0                     Satterthwaite's degrees of freedom =  37.3427

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.5327         Pr(|T| > |t|) = 0.9347          Pr(T > t) = 0.4673
```

Appendix 6.2. T-test of Profitability

```
. keep if SalesOutcome == 1
(304 observations deleted)

. ttest Profit, by (Region) unequal

Two-sample t test with unequal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
   India |      17    48.64706    2.799608    11.54308    42.71216    54.58196
      UK |     267    46.22846    .5663574    9.254356    45.11335    47.34358
---------+--------------------------------------------------------------------
Combined |     284    46.37324    .5577936    9.400105    45.27529    47.47119
---------+--------------------------------------------------------------------
    diff |            2.418594     2.85632               -3.598864    8.436053
------------------------------------------------------------------------------
    diff = mean(India) - mean(UK)                                 t =   0.8468
H0: diff = 0                     Satterthwaite's degrees of freedom =  17.3346

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.7957         Pr(|T| > |t|) = 0.4087          Pr(T > t) = 0.2043
```

Appendix 6.3. T-test of Customer Size

```
. ttest ProfitofCustomer, by (Region) unequal

Two-sample t test with unequal variances
```

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
India	35	1.082	.0779062	.4608994	.9236755	1.240324
UK	553	1.012007	.0219528	.5162407	.968886	1.055128
Combined	588	1.016173	.0211567	.5130234	.9746214	1.057726
diff		.0699928	.0809401		-.093645	.2336306

```
    diff = mean(India) - mean(UK)                                 t =   0.8647
H0: diff = 0                     Satterthwaite's degrees of freedom =  39.5984

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.8038         Pr(|T| > |t|) = 0.3924          Pr(T > t) = 0.1962
```

Appendix 6.4. T-test of Conversion rate

```
. ttest SalesOutcome, by (Region) unequal

Two-sample t test with unequal variances
```

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
India	35	.4857143	.0857143	.5070926	.3115219	.6599067
UK	553	.482821	.0212688	.5001572	.4410432	.5245988
Combined	588	.4829932	.0206253	.5001362	.4424849	.5235015
diff		.0028933	.0883137		-.1758413	.181628

```
    diff = mean(India) - mean(UK)                                 t =   0.0328
H0: diff = 0                     Satterthwaite's degrees of freedom =  38.3068

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.5130         Pr(|T| > |t|) = 0.9740          Pr(T > t) = 0.4870
```

The “mean” column represents the conversion rate, since the outcome variables are 1 and 0.

Appendix 7.1. Chi-square Test of Sales Outcome by Region

```
. tabulate Region SalesOutcome, chi2
```

Region	Sales Outcome 0	1	Total
Africa	38	55	93
Americas	49	55	104
Canada	2	4	6
India	18	17	35
Japan	6	10	16
Other Europe	92	66	158
Singapore	17	6	23
Spain	11	1	12
UK	286	267	553
Total	519	481	1,000

```
          Pearson chi2(8) =  22.2626   Pr = 0.004
```

Appendix 7.2. ANOVA of Sales Value across Regions

```
. oneway SalesValueinM~n Region, tabulate

             |   Summary of Sales Value in Million
      Region |        Mean   Std. dev.       Freq.
-------------+-------------------------------------
      Africa |   7.8188172   1.8945533          93
    Americas |   7.7584615   1.8853378         104
      Canada |       8.185   1.2747353           6
       India |   8.1868571   2.3094636          35
       Japan |    8.900625   2.0140356          16
Other Eur.. |   7.8414557   1.8813211         158
   Singapore |   8.1969565   2.1818412          23
       Spain |   7.9616667   2.0210476          12
          UK |   8.1538698   2.0128937         553
-------------+-------------------------------------
       Total |      8.0442   1.9829336       1,000

                        Analysis of variance
    Source              SS         df      MS            F     Prob > F
------------------------------------------------------------------------
Between groups      39.5463008      8   4.9432876      1.26     0.2610
 Within groups      3888.54746    991   3.92386222
------------------------------------------------------------------------
    Total           3928.09376    999   3.93202579

Bartlett's equal-variances test: chi2(8) =   5.5430    Prob>chi2 = 0.698
```

Appendix 7.3. Summary of Conversion Rate, Sales Value, and Sample Size by Region

```
. list Region Conversion SalesValue N, noobs
```

Region	Convers~n	SalesVa~e	N
Canada	.66666667	8.185	6
Japan	.625	8.900625	16
Africa	.59139785	7.8188172	93
Americas	.52884615	7.7584615	104
India	.48571429	8.1868571	35
UK	.48282098	8.1538698	553
Other Europe	.41772152	7.8414557	158
Singapore	.26086957	8.1969565	23
Spain	.08333333	7.9616667	12

Appendix 7.4. Chi-square Test of Independence between Product and Sales Outcome

```
. tabulate Product SalesOutcome, chi2
```

Product	Sales Outcome 0	1	Total
ContactSys	14	6	20
Finsys	83	34	117
GTMSys	236	227	463
LearnSys	55	71	126
Lifesys	58	54	112
Logissys	9	20	29
Procsys	64	69	133
Total	519	481	1,000

Pearson chi2(6) = 29.0292 Pr = 0.000

Appendix 7.5. Win Rates by Product and Region

```
. list prod_region wins total winrate, sep(0)
```

	prod_region	wins	total	winrate
1.	ContactSys_UK	6	20	.3
2.	Finsys_Other Europe	15	38	.39473684
3.	Finsys_UK	19	79	.24050633
4.	GTMSys_Africa	38	62	.61290323
5.	GTMSys_Americas	50	91	.54945055
6.	GTMSys_Japan	3	9	.33333333
7.	GTMSys_Other Europe	28	56	.5
8.	GTMSys_Singapore	6	23	.26086957
9.	GTMSys_Spain	1	12	.08333333
10.	GTMSys_UK	101	210	.48095238
11.	LearnSys_Africa	17	31	.5483871
12.	LearnSys_UK	54	95	.56842105
13.	Lifesys_Canada	1	1	1
14.	Lifesys_Other Europe	23	64	.359375
15.	Lifesys_UK	30	47	.63829787
16.	Logissys_UK	20	29	.68965517
17.	Procsys_Americas	5	13	.38461538
18.	Procsys_Canada	3	5	.6
19.	Procsys_India	17	35	.48571429
20.	Procsys_Japan	7	7	1
21.	Procsys_UK	37	73	.50684932

Appendix 8.1. Z-test of conversion rate: WSES > 50% vs ≤ 50% ownership

```
Two-sample test of proportions                   WSES ≤ 50%: Number of obs =      169
                                                 WSES > 50%: Number of obs =      831

       Group |       Mean   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
  WSES ≤ 50% |   .3964497   .0376277                      .3227008    .4701986
  WSES > 50% |   .4981949   .0173447                         .4642    .5321899
-------------+----------------------------------------------------------------
        diff |  -.1017452   .0414328                     -.1829521   -.0205384
             |  under H0:   .0421611   -2.41   0.016
------------------------------------------------------------------------------
        diff = prop(WSES ≤ 50%) - prop(WSES > 50%)                z =  -2.4132
    H0: diff = 0

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(Z < z) = 0.0079         Pr(|Z| > |z|) = 0.0158          Pr(Z > z) = 0.9921
```

Appendix 9 Chi-Squared Test of Sales Conversion Rates between E/V and F/L Lead Groups

observed

	win	lost	total	
E/V	388	400	788	0.788
F/L	93	119	212	0.212
	481	519	1000	
	0.481	0.519		

expected

	win	lost	total
E/V	379.028	408.972	788
F/L	101.972	110.028	212
	481	519	1000

chi squared p-value **0.164736**

Appendix 10 Scatter plot of marketing expenses and profitability